February 1990, after a full year of false starts and mistakes, our calculations coalesced and gave an answer.

I was surprised and shocked. Despite the wormhole's attempt to defocus them, the vacuum fluctuations tended to refocus of their own accord (Figure 14.11). Defocused by the wormhole, they splayed out from the mouth on Earth as though they were going to miss the spacecraft; then of their own accord, as though being attracted by some mysterious force, they zeroed in on the wormhole mouth in Carolee's spacecraft. Returning to Earth through the wormhole, they then splayed out from the mouth on Earth again, and zeroed in once again on the mouth in the spacecraft. Over and over again they repeated this motion, building up an intense beam of fluctuational energy.

Will this beam of electromagnetic vacuum fluctuations be intense enough to destroy the wormhole? Kim and I asked ourselves. For eight months, February to September 1990, we struggled with this question. Finally, after several flip-flops, we concluded (incorrectly) "probably not." Our reasoning seemed compelling to us and to the several colleagues we ran it past, so we laid it out in a manuscript and submitted it to the *Physical Review.*

Our reasoning was this: Our calculations had shown that the circulating electromagnetic vacuum fluctuations are *infinitely intense only for a vanishingly short period of time.* They rise to their peak at precisely the instant when it is first possible to use the wormhole for backward time travel (that is, at the moment when the wormhole first becomes a time machine), and then they immediately start to die out; see Figure 14.12.

Now, the (ill-understood) laws of quantum gravity seem to insist that there is no such thing as a "vanishingly short period of time." Rather, just as fluctuations of spacetime curvature make the concept of length meaningless on scales smaller than the Planck–Wheeler length, 10^{-33} centimeter (Figure 14.3 and associated discussion), so also the curvature fluctuations should make the concept of time meaningless on scales smaller than 10^{-43} second (the "Planck-Wheeler time," which is equal to the Planck–Wheeler length divided by the speed of light). Time intervals shorter than this cannot exist, the laws of quantum gravity seem to insist. The concepts of *before* and *after* and *evolution with time* make no sense during intervals so small.